ENGINEERING DRAFT — NOT FOR FABRICATION — See hardware/omi-mk-d1/README.md

01 MCU / RTC / UI

U1
STM32U535CEU6
VDD
VBAT
PC13
PC14
PC15
PH0
PH1
NRST
VDDA
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB2
PA13
PA14
VCAP
VSS
VDD
PB12
PB15
PB14
PB13
PA8
PA9
PA10
PA11
PA12
PB3
PB4
PB5
PB6
PB7
PB8
PB9
VSS
VDD
PA15
PC10
PC11
PC12
PD2
PB10
PB11
VSS
VDD
BUTTON_N
CHG_N
STATUS_LED
USB_DM_MCU
USB_DP_MCU
SWDIO
GND
VIN3
SWCLK
PDM_CLK_MCU
PDM_DATA
BOOT0
GND
VIN3
GND
C1
100n
C4
10u
C9
100n
R2
10k
Y1
32.768kHz CL=7pF TBD
C11
10p tentative
C12
10p tentative
J7
TC2030 SWD target pads
SWCLK
SWDIO
NRST
SWD_present
SW1
B3U-1000P (candidate)
BUTTON_N
R4
100k
R5
3.3k
LED_A
D1
red LED
C2
100n
C3
100n
C5
4.7u
C6
100n
C7
1u
R1
0R
R3
100k
BOOT0

02 USB-C / CHARGER

J1
USB4135-GF-A (candidate)
GND
VBUS
CC1
USB_DP
USB_DM
SBU1
VBUS
GND
GND
A1
A4
A5
A6
A7
A8
B1
B4
B5
B6
B7
B8
SHIELD
VBUS
CC2
DN
DP
SBU2
VBUS
GND
GND
B9
B12
S1
S2
S3
S4
R6
5.1k 1%
CC1
R7
5.1k 1%
CC2
R8
0R tuning
USB_DP
USB_DP_MCU
R9
0R tuning
USB_DM
USB_DM_MCU
U2
USBLC6-2P6
IO1
GND
IO2
I/O1
VBUS
I/O2
C13
2.2u 10V
VBUS
U3
BQ25101YFPR
VBUS
BAT
ISET
IN
OUT
ISET
TS
CHG
VSS
BAT_TS
CHG_N
GND
R10
6.19k 1%
ISET
R11
100k
VIN3
CHG_N
C14
2.2u
BAT
J2
SM02B-SRSS-TB / factory protected round 150mAh pack TBD
PACK+
PACK-
TH1
NCP15XH103F03RC 10k NTC
BAT_TS
Q2
2N7002 (charge inhibit)
CHG_DISABLE
G
D
S
BAT_TS
R12
100k
CHG_DISABLE
Q3
2N7002 (VBUS detect)
VBUS_GATE
GND
G
D
S
VBUS_N
R13
100k
VBUS
VBUS_GATE
R14
1M
VBUS_GATE
R15
100k
VIN3
VBUS_N
#FLG01
PWR_FLAG
GND
pwr
#FLG02
PWR_FLAG
VDDA
pwr
#FLG03
PWR_FLAG
VBUS
pwr
#FLG04
PWR_FLAG
SYS
pwr

03 POWER / FUEL GAUGE

Q2
BAT54WS (candidate: leakage/forward drop review pending)
VBUS
A
K
SYS
Q1
DMP2035U (candidate: power path review pending)
VBUS
G
S
D
BAT
SYS
R16
100k
VBUS
C15
4.7u
SYS
U4
TPS7A0233PDBVR
SYS
IN
GND
EN
NC
OUT
VIN3
U5
TPS7A0218PDBVR
VIN3
IN
GND
EN
NC
OUT
VCAM
C16
1u
VIN3
C18
1u
VCAM
C19
1u
VIN3
R17
100k
MIC_EN
TP1
TestPoint
BAT
TP2
TestPoint
SYS
TP3
TestPoint
VIN3
TP4
TestPoint
VCAM
TP5
TestPoint
SD_V3
TP6
TestPoint
GND
R33
1M 1%
BAT
BAT_ADC
R34
1M 1%
BAT_ADC
GND
C28
10n
BAT_ADC
GND

04 AUDIO / SD / INDICATORS

U8
TS4230
MIC_DATA_CAM
GND
GND
MIC_MASC_CAM
DATA
SELECT
GND
WAKE
THOEL
CLK
VDD
MIC_THOEL_CAM
MIC_CLK_CAM
VCAM
R19
TXS0104ERUTR
VCAM
MIC_MASC_CAM
MIC_THOEL_CAM
MIC_DATA_CAM
MIC_CLK_CAM
GND
VCCA
A1
A2
A3
A4
OE
B4
B3
B2
B1
VCAM
PDM_CLK
PDM_DATA
MIC_THOEL
MIC_MASC
VIN3
R21
22R tentative
PDM_CLK_MCU
PDM_CLK
C22
100n
VCAM
C23
100n
VCAM
C24
100n
VIN3
U10
TPS22918DCYFFR
VIN3
SD_EN
ON
VOUT
SD_EN
GND
R22
100k
SD_EN
C25
1u
VIN3
U11
CSNP32GM01-BCW
SD_CS_N
SD_CLK
SD_CLK
GND
DI
VSS
DO
SCK
VCC
SD_MOSI
SD_MISO
SD_CLK
SD_V3
R23
22R tentative
SD_CLK_MCU
SD_CLK
R24
47k
SD_V3
SD_MOSI
R25
47k
SD_V3
SD_MISO
R28
47k
SD_V3
SD_CS_N
C26
100n
SD_V3
C27
22u effective TBD
SD_V3
R32
4.8k tentative
STATUS_LED
STATUS_LED_A
D6
amber LED 0402 TBD
STATUS_LED_A
A
K

Unrouted PCB / no manufacturing release

Sheet: /
File: omi-mk-d1.kicad_sch

Title: Pendant Recorder — Omi-derived electrical design

Size: A2	Date: 2026-10-24	Rev: omi-mk-d1 DRAFT
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